

Research Paper

Structure-Based Virtual Screening and Molecular Dynamics Simulation of FDA-Approved Drugs Targeting FGFR3 K650E Mutation in Bladder Cancer

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ABSTRACT

Bladder cancer is one of the most common malignancies of the urinary tract. The most-preferred alterations in bladder cancer include mutations in fibroblast growth factor receptor 3 (FGFR3), primarily the K650E substitution which induces constant activation of FGFR3 kinases and oncogenic signaling. In this study, we are attempting to employ a broad in silico approach combining structure-based virtual screening together with molecular dynamics simulations to repurpose potential FDA-approved drugs for inhibiting the FGFR3 K650E mutant protein. The 3D crystal structure of the mutant (PDB ID: 4K33) was refined and validated using PDB-REDO and SAVES v6.0 tools in order to ensure that the model is geometrically reliable. The FDA-approved compound library was screened through DrugRep, where Irinotecan, Eltrombopag, and Lurasidone came through as top candidates based on docking scores. The protein-ligand complexes were further assessed through normal mode analysis with the iMODS server to study their conformational flexibility along with binding site characteristics and dynamic behavior. The analysis revealed crucial interaction residues along with dynamic hinge regions that affirm strong and biologically plausible molecular binding. The results support a platform for repurposing existing drugs against the FGFR3 K650E variant and provide computational grounds for future experimental validation.

1. Introduction

Bladder cancer poses serious challenges to global health. It is ranked among the most common cancers affecting the urinary tract [1]. Bladder cancer can be presented as

non-muscle invasive or muscle invasive forms, the latter of which is more aggressive and harder to treat [2]. The pathogenesis of bladder cancer represents a

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series of interacting factors that incorporate environmental influences and genetic alterations, which contribute to tumor initiation, progression, and recurrence [3]. One of the most frequently mutated genes in bladder cancer is fibroblast growth factor receptor 3 (FGFR3), a transmembrane receptor tyrosine kinase involved in cellular proliferation, differentiation, and survival [4]. FGFR3 mutation also leads to constitutive activation of its kinase domain and subsequent sustained oncogenic signals [5].

Among the mutations that have been reported, the K650E substitution in the FGFR3 kinase domain is of great importance as it conveys gain of function [6]. Activation of the receptor in a ligand-independent manner through this mutation has been proposed in association with many cancers, including Thanatophoric Dysplasia Type II, Spermatocytic Seminoma, and more aggressive forms of bladder cancer [7]. The crystal structure of FGFR3 K650E mutation (PDB ID: 4K33) is thus a beneficial target for silico drug designing of this oncogenic variant [8]. Therefore, targeting mutant FGFR3 never been more appealing owing to its role in tumor progression along with therapy resistance [9].

In the past few years, structure-based drug repurposing has garnered attention as a rational strategy to expedite cancer drug discovery [10]. Drug repurposing entails the process of identifying new therapeutic applications for drugs that have already received FDA approval and offered a quicker path to candidate introduction to clinical testing by using pre-existing pharmacological information and characteristics [11]. Existing drugs are evaluated for their capability to bind to and inhibit targets specific to the disease using a variety of advanced techniques including molecular docking, virtual screening, and molecular dynamics simulations [12]. When applied in a cancer context to proteins such as FGFR3, it provides for efficient exploration towards potential new therapeutics [13].

This study aims to obtain candidate FDA-approved drugs characterized by strong affinities for binding directly to K650E mutant FGFR3 protein implicated in chronic bladder cancer [14]. The workflow constituted retrieval of the 3D structure of the mutated protein (PDB ID: 4K33), refinement and validation for its accuracy [15], and receptor-based virtual screening on DrugRep [16]. Selected best candidate compounds then underwent normal mode analysis (NMA) using the iMODS server to

understand the conformational dynamics and stability of the protein-ligand complexes [17]. This integrated computational approach is designed to highlight potential candidates for experimental validation and downstream profiling against this target [18].

2. Materials and Methods

2.1 Target Selection and Preparation

This study has FGFR3 chosen as a target protein with gain-of-function mutation K650E postulated to have a role in

Thanatophoric Dysplasia Type II, spermatocytic seminoma, and aggressive forms of bladder cancer [19]. The 3D structure relating to this mutant kinase domain is available in the Protein Data Bank as PDB ID: 4K33. The said structure in .pdb format is retrieved from the RCSB PDB portal [20]. The next step is the download of the .pdb format of the mutant kinase by the RCSB PDB (<https://www.rcsb.org/>) for in-silico analysis [21].



Figure 2.1: Crystal structure of human K650E mutant in complex with GDP

To realize high-quality structural input for the docking studies, the protein has been refined through use of the PDB-REDO server which optimizes crystallographic structures by re-refining atomic positions and B-factors [22]. The post-refinement structure was subsequently validated with

the help of the SAVES v6.0 server that contains various quality assessment tools such as the Ramachandran plot (via PROCHECK), ERRAT, VERIFY3D, and PROVE [23]. This validation is meant to ensure the geometry and energy reliability

of the protein structure for computational studies downstream [24].

2.2 Virtual Screening

The refined FGFR3-K650E structure, having passed through structural validation, was uploaded to the DrugRep webserver, which performs receptor-based virtual screening of FDA-approved drugs [25]. DrugRep uses docking algorithms to estimate the binding affinity of drug molecules to a given protein target [26]. An exhaustive screening was thus performed on the FDA-approved compounds, and results were ranked on the basis of docking scores (in kcal/mol). Compounds with more negative scores were postulated for stronger binding affinities [27]. Top 100 compounds based on their docking performances were selected and tabulated with DrugBank IDs, drug names, and docking scores. Interestingly, Irinotecan (DB00762) had the best binding scoring with -10.8 kcal/mol, with Eltrombopag (-10.4), Lurasidone (-10.2), and others following [28].

2.3 Protein-Ligand Complex Preparation

Then the protein-ligand complexes were fashioned for further consideration from virtual screening results with the top-ranked compounds. A special focus was given to

the top hits such as Irinotecan and Eltrombopag [29]. Each ligand from the docking poses was docked into the active site of the FGFR3 mutant obtained from DrugRep [30]. These complexes were saved as PDB files for further molecular dynamics simulation. This step is vital to evaluate not only the static binding affinity but also the dynamic stability and flexibility of the protein-ligand interactions [31].

2.4 Molecular Dynamics Simulation

Conformational dynamics and structural flexibility of protein-ligand complexes can be studied using Normal Mode Analysis (NMA) with the help of the iMODS server. iMODS is a primary tool for rapid assessment of macromolecular dynamics on the basis of internal coordinates [32]. Herein, protein-ligand complexes were uploaded and scored using several important parameters. These included molecular deformability, which indicates the flexibility of residues; B-factors, which reflect atomic displacement; eigenvalues, representing the stiffness of the system; variance, associated with the magnitude of atomic motion; and the covariance matrix, which shows the correlation of movements between residues [33].

3. Results



3.1 Protein structure Validation

3.1.1. PDB REDO

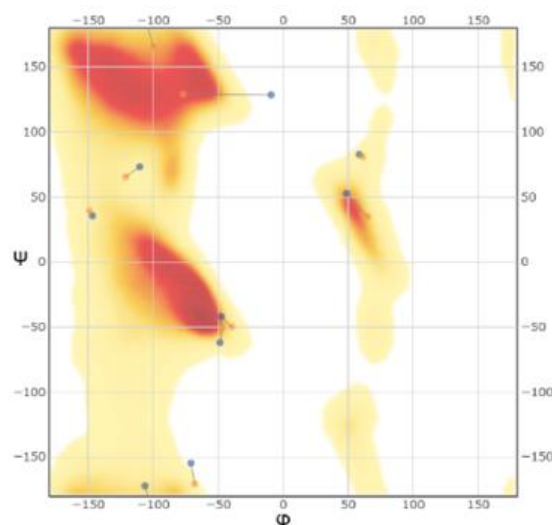


Figure 3.1: Backbone Conformational Analysis of Refined Mutant Androgen Receptor Using Kleywegt-like Plot

The obtained Kleywegt-like plot relating to the refined FGFR3 protein structure visually creates an assessment of the backbone dihedral angles (ϕ and ψ) of the amino acid residues with respect to statistically favourable regions for high-quality protein structures. The majority of the residues (represented by blue dots) lie in the highly populated red areas, meaning, energetically favourable and sterically permitted conformations. This distribution indicates that most of the backbone torsion angles in the protein match with the ideal

geometry, suggesting a structurally sound and well-folded model. A few residues can be viewed in the yellow or unshaded parts - assumed to be unfavorable or forbidden regions; nevertheless, few of them might be those which correspond to flexible loop portions or functionally significant areas near the active or mutation sites like K650E mutation. Overall, the plot attests for the integrity of stereochemical structure and approval for the model for further computational studies like virtual screening and molecular dynamics simulation.

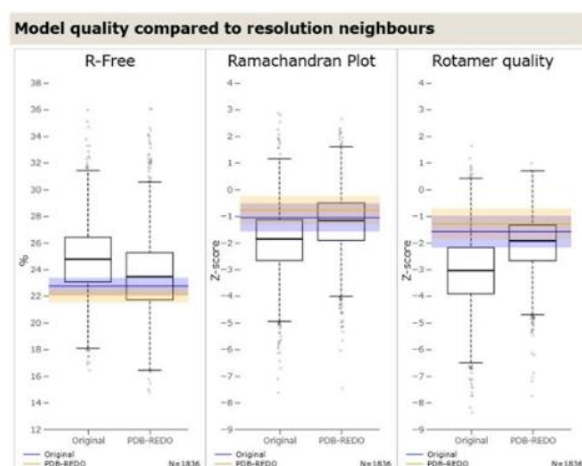


Figure 3.2: Improved Structural Validation of Mutant Androgen Receptor via PDB-REDO: A Comparative Z-Score and R-Free Assessment

The comparative image shows model quality analysis between original PDB structures and their PDB-REDO refined versions by means of three key validation metrics: R-Free percentage, Ramachandran plot Z-score, and rotamer quality Z-score. Box plots for original and PDB-REDO models are shown side by side for each panel, along with shaded bands indicating the expected quality range for structures of similar resolution. The first panel indicates that the R-Free values, a measure of model accuracy with regard to experimental data, are slightly lower in the PDB-REDO models, suggesting a better fit from model to data. The median R-Free is lowered while refined ones bunch closer to their expected values for the resolution. The second panel implicates higher-than-original scores (better) for the PDB-REDO

set Ramachandran plot Z-scores that indicate backbone geometry quality, whereas remaining outliers count seems to be significantly dropped amongst the models. The same third panel Z-score comparison that interrogates side-chain conformations shows improvement in refined models with higher medians that coincide with the expected quality range. All-in-all, the data in the figure show that PDB-REDO refinement leads to measurable improvements in the model quality metrics and thus moves the structure closer to their perfect geometric and statistical idea. This validates the further use of fully automated refinement programs, like PDB-REDO, for increasing crystallography model reliability.

Table 3.1: Comparative Validation Metrics of Original and PDB-REDO Refined Structures

Metric	Original	PDB-REDO
Crystallographic Refinement		
R	0.1895	0.1785
R-free	0.2351	0.2214
Bond length RMS Z-score	0.853	0.368
Bond angle RMS Z-score	0.795	0.561
Model Quality (Percentiles)		
Ramachandran plot normality	52	60
Rotamer normality	66	72
Coarse packing	67	65
Fine packing	77	82
Bump severity	28	82
Hydrogen bond satisfaction	77	73

The table compares the validation metrics of the original structural model with its PDB-REDO-refined structure, focusing aspects of crystallographic refinement and model quality. In respect to the crystallographic refinement, there is considerable enhancement in the model's correspondence with experimental data. The R-value decreased from 0.1895 to the lower value of 0.1785, alongside the decline of the R-free value from 0.2351 to 0.2214, indicative of a better overall fit and cross-validation. Besides, bond length RMS Z score improved significantly by 0.853 to 0.368, while the bond angle RMS Z score was also lower at 0.795 than after refinement, which yielded a score of 0.561.

These are reductions which reflect improvement in the geometrical accuracy since a refined model shows fewer deviations from ideal bond lengths and angles. In the model quality section, most parameters perked up with the refined structure with regard to their corresponding benchmarks.

The Ramachandran plot normality grew from 52 to 60, and rotamer normality ran from 66 to 72, indicating better conformations in backbone and side chains. It further improved fine packing from 77 to 82, with thus more favourable local atomic environments. However, bump severity strikingly increased from 28 to 82, a

reflection of the considerable reduction in steric clashes and a cleaner construction of the model that is also more physically realistic. Fine packing shows a slight decline (from 67 to 65) in coarse packing and hydrogen bond satisfaction (from 77 to 73). However, these minor declines do not add up to the overall considerable gains in

model quality and accuracy. Summarily, PDB-REDO refinement has clearly improved both the structural and stereochemical quality of the model efficiency in optimizing protein structure for better reliability and interpretability.

3.1.2. Save v6.1 Result

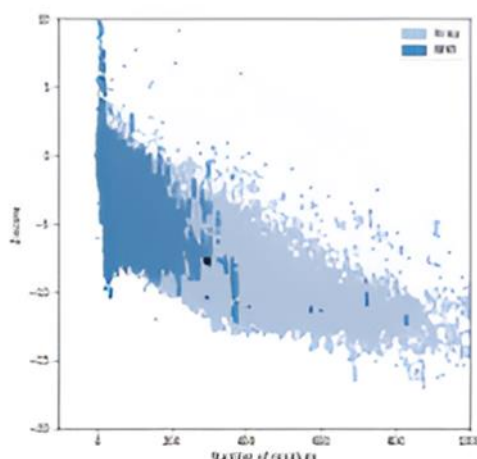


Figure 3.3: Detection of Structural Outliers: Elevated Z-Score Reveals Anomalous Model Geometry in Mutant Androgen Receptor (overall model quality z score -7.10)

The scatter plot correlating Z-score and resolution provides a large-scale view of model quality-an approach traditionally employed by structural measured biologists to ascertain the reliability of macromolecular models. On the x-axis, the resolution is in angstroms (Å) with lower values indicating higher resolution and thereby a more detailed structural representation. On the y-axis are placed Z-

scores, providing a numerical value on how much the quality of a given model deviates from the expected standards at that resolution. A Z-score of near zero would mean generic model quality, while increasingly negative values would suggest poorer structural accuracy.

The points on the plot show data for both the original PDB models and PDB-REDO models and different shades of blue

represent the models. The distribution of points across the plot generally shows that model quality decreases as resolution worsens. On this background, the current model score, with a Z-score percentile of 47.03, is slightly colored-around the middle when compared to other structures with similar resolution meaning an average level of quality-neither much better nor worse

than its peers. The model can be said to have average quality and would benefit from some improvements to lift it above median percentile. This kind of assessment helps with deciding whether structural accuracy of the model would benefit from further refinements or some manual corrections

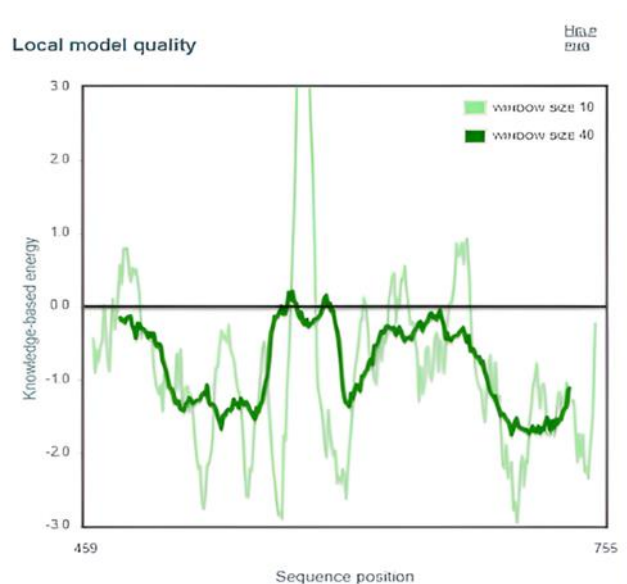


Figure 3.4: Assessment of Local Structural Reliability: Knowledge-Based Energy Profile of Androgen Receptor Variant"

A graph entitled "Local Model Quality" measures the reliability of a protein structure through its amino acid sequence from position 459 to position 755, with knowledge-based energy values on the y axis-the lower (more negative) the value, the better the local model quality. There are two curves here, one representing a window

size of 10 in light green along with one for a window size of 40 in dark green. Whereas the light green curve varies significantly to capture fine-grained local variations, the dark green speaks to the more robust quality trends of the model. Most of the sequence lies below the neutral energy line at $y = 0$, indicating good quality for the model.

However, certain extreme regions—some near the ends of the sequence and one sharp spike in the middle line—really cross this threshold; hence the possible local structural inaccuracies or misfolding would appear. The different window sizes give complementary information about the model: a smaller window addressing

specific residues with problems while a larger window giving a cleaner view of regional consistency. Identifying these strong and weak areas in the model is important when interpreting functional significance or guiding model refinement.

3.3. Results of Virtual Screening Outcomes

Table 3.2: Ranked List of Drug Compounds by Binding Affinity Scores

S.No	DrugBank ID	Name	Score	S.No	DrugBank ID	Name	Score
1	DB00762	Irinotecan	-10.8	51	DB00932	Tipranavir	-8.8
2	DB06210	Eltrombopag	-10.4	52	DB01238	Aripiprazole	-8.8
3	DB08815	Lurasidone	-10.2	53	DB00696	Ergotamine	-8.8
4	DB01232	Saquinavir	-10.1	54	DB06144	Sertindole	-8.8
5	DB06212	Tolvaptan	-10.0	55	DB09209	Pholcodine	-8.8
6	DB08827	Lomitapide	-10.0	56	DB11730	Ribociclib	-8.7
7	DB00941	Hexafluronium	-9.9	57	DB12001	Abemaciclib	-8.7
8	DB09195	Loripirazole	-9.8	58	DB09074	Olaparib	-8.7
9	DB11652	Tucatinib	-9.8	59	DB06626	Axitinib	-8.7
10	DB09280	Lumacaftor	-9.8	60	DB00303	Ertapenem	-8.6
11	DB00398	Sorafenib	-9.7	61	DB11791	Capmatinib	-8.6
12	DB00734	Risperidone	-9.6	62	DB11596	Levoleucovorin	-8.6
13	DB12978	Pexidartinib	-9.6	63	DB00872	Conivaptan	-8.6
14	DB13766	Lidoflazine	-9.6	64	DB00950	Fexofenadine	-8.6
15	DB00246	Ziprasidone	-9.6	65	DB00490	Buspirone	-8.5
16	DB11855	Revefenacin	-9.5	66	DB00251	Terconazole	-8.5
17	DB04908	Flibanserin	-9.5	67	DB12141	Gilteritinib	-8.5
18	DB01100	Pimozide	-9.5	68	DB09076	Umeclidinium	-8.5
19	DB00137	Lutein	-9.4	69	DB00984	Nandrolone	-8.5

20	DB00511	Acetyldigitoxin	-9.4	70	DB11611	Lifitegrast	-8.5
21	DB01267	Paliperidone	-9.4	71	DB08881	Vemurafenib	-8.5
22	DB06809	Plerixafor	-9.4	72	DB08899	Enzalutamide	-8.5
23	DB09128	Brexiprazole	-9.4	73	DB00590	Doxazosin	-8.5
24	DB09102	Daclatasvir	-9.3	74	DB11703	Acalabrutinib	-8.5
25	DB13931	Netarsudil	-9.3	75	DB00450	Droperidol	-8.4
26	DB11637	Delamanid	-9.3	76	DB13954	Estradiol	-8.4
27	DB12877	Oxatomide	-9.3	77	DB08995	Diosmin	-8.4
28	DB00342	Terfenadine	-9.3	78	DB05039	Indacaterol	-8.4
29	DB12020	Tecovirimat	-9.3	79	DB12371	Siponimod	-8.4
30	DB06684	Vilazodone	-9.2	80	DB08439	Parecoxib	-8.4
31	DB06755	Beta carotene	-9.2	81	DB11256	Levomefolic	-8.4
32	DB08896	Regorafenib	-9.2	82	DB14541	Hydrocortisone	-8.4
33	DB12887	Tazemetostat	-9.1	83	DB06817	Raltegravir	-8.4
34	DB01396	Digitoxin	-9.1	84	DB11942	Selinexor	-8.3
35	DB00222	Glimepiride	-9.1	85	DB08893	Mirabegron	-8.3
36	DB01184	Domperidone	-9.1	86	DB11901	Apalutamide	-8.3
37	DB04861	Nebivolol	-9.0	87	DB00650	Leucovorin	-8.3
38	DB00878	Chlorhexidine	-9.0	88	DB13074	Macimorelin	-8.3
39	DB04868	Nilotinib	-9.0	89	DB00673	Aprepitant	-8.2
40	DB06077	Lumateperone	-9.0	90	DB00894	Testolactone	-8.2
41	DB11742	Ebastine	-9.0	91	DB01003	Cromoglicic	-8.1
42	DB01167	Itraconazole	-9.0	92	DB00843	Donepezil	-7.9
43	DB09374	Indocyanine	-9.0	93	DB01261	Sitagliptin	-7.8
44	DB04918	Ceftobiprole	-8.9	94	DB00836	Loperamide	-7.7
45	DB01126	Dutasteride	-8.9	95	DB11978	Glasdegib	-7.0
46	DB13520	Metergoline	-8.9	96	DB15233	Edoxaban	-7.0
47	DB06016	Cariprazine	-8.9	97	DB00206	Pralatrexate	-7.0
48	DB08912	Dabrafenib	-8.9	98	DB01126	Glipizide	-7.0
49	DB01259	Lapatinib	-8.8	99	DB06813	Avapritinib	-7.0
50	DB06695	Dabigatran	-8.8	100	DB15233	Reserpine	-7.0

The dataset features 100 compounds pertaining to their binding scores, extremely well representing predicted affinities to a certain target while performing a molecular docking study. It has binding scores from -10.8 to -7.0, and it typically indicates that the more negative, the stronger binding is predicted. The first one in the long list of performance compounds is Irinotecan with a score of -10.8; it is followed by Eltrombopag (-10.4), Lurasidone (-10.2), and Saquinavir (-10.1), which suggests these molecules really hold great potential to bind effectively at the target. Overall, significant numbers of compounds, especially those scoring less than -9.5, are very much indicated to have strong binding affinities, making them candidates for further experimental validation.

Most compounds lie somewhere between the scores of -9.4 and -8.5, showing that they have a moderate to high binding affinity and hence seem to be promising chemical spaces for repurposing or lead optimization. Compounds in the lower end of the scale-from scores of -8.4 to -7.0-reflect weak binding and might not be

considered for further studies unless they offer other pharmacological gain. Among other things, the dataset can be classified into an array of therapeutic categories such as anticancer drugs (e.g. Dabrafenib, Regorafenib), antivirals (e.g., Saquinavir, Daclatasvir), antipsychotics (e.g., Risperidone, Ziprasidone), and cardiovascular drugs (e.g., Nilotinib, Nebivolol), which open up avenues for the repurposing of drugs across different therapeutic areas.

However, the above is not really true about the table. For instance, DrugBank ID DB01126 is repeated, once because of association with Dutasteride and the other with Glipizide; it denotes a probable error in labeling or entry. Similar is the case with DB15233; it is attributed to both Edoxaban and Reserpine. These corrections should be carried out to ensure data accuracy and to avoid confusion during further analysis. To this end, it is a dataset that opens many dosages on foresighted drug-target interaction and is a strong basis for further in vitro or in silico studies mainly focusing on the top-scoring compounds with strong predicted affinities.

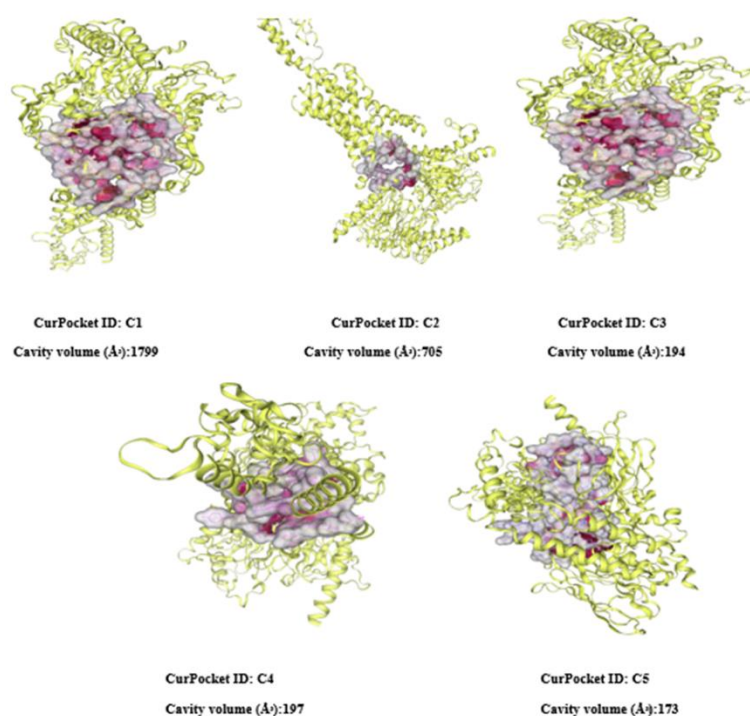


Figure 3.5: Predicted Binding Pockets and Cavity Volumes in Androgen Receptor Variant Structure

Here we have an image along with 5 binding cavities found between protein and ligand that has been identified using a tool named CurPocket. It is specifically meant for the detection and visualization of potential ligand-binding sites on protein surfaces. These are displayed by structures in ribbon format (in yellow) with the binding pockets highlighted in a semi-transparent pink/purple surface and possible active regions shown in red. These cavities, C1 to C5, show variability in volume which in turn has uplifting importance in terms of how well they could house potential ligands. Largest among them, C1 is able to boast of a cavity volume of 1799 Å³ thereby making itself the most spacious and potentially the best site to bind

large or complicated ligands. This kind of volume suggests that C1 is possibly a very druggable site well situated for structure-based drug design approaches requiring significant spaces with diverse molecular scaffolds. On the contrary, this small volume difference makes C2, which alone has a volume of 705 Å³, smaller compared to C1, but it is still sufficiently large to accept medium sized ligands suitable to give flexibility with specificity. The left of the cavities-C3, C4, and C5-is pretty much smaller. Their volumes in Å³ ranges between 173 and 197. These smaller pockets are believed to be more selective and suitable for fragment-based drug discovery which is an approach in which very small molecular fragments are

designed with respect to fit into small constrained binding spaces. In terms of their structures, the pockets have different depths and locations: while some are apparently deeply buried in the protein, others are more surface oriented, which can

have a bearing on ligand accessibility and binding affinity. Altogether, cave diversity in size and position leaves several openings for drug discovery, with C1 standing as the most compelling target due to its substantial, accessible binding volume.

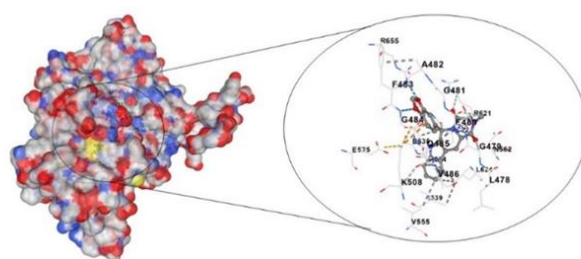


Figure 3.6: Molecular Interaction Mapping of Ligand within Protein Binding Pocket

This image shows a full molecular representation of a protein-ligand interaction as applied in structural biology and drug development studies. On the left, the space-filling surface model of the protein is shown with areas colored according to electrostatic surface potential; red for negative, blue for positive, and neutral or hydrophobic regions colored white. The binding pocket or active site of the protein is highlighted with an oval black marker. To the right is a zoomed-in view of this active site, showing specific interactions between the ligand (as shown in stick representation) and surrounding amino acid residues. Residues, such as R655, E575, and K508, participate in

hydrogen bonding and electrostatic interactions with the ligand, with the dashed lines indicating such interactions. The signature of polar, charged, and aromatic residues with glycine, arginine, lysine, glutamate, and phenylalanine strongly suggests that it is a mixture of electrostatic and hydrophobic interactions that forms the basis of ligand stabilization within the pocket. Such a study would be pertinent to structure-based drug designing, enzyme engineering, and molecular recognition processes, allowing researchers to identify crucial binding interactions and design effective drugs or proteins with modified binding specificity.

3.4. Molecular dynamic stimulation analysis.

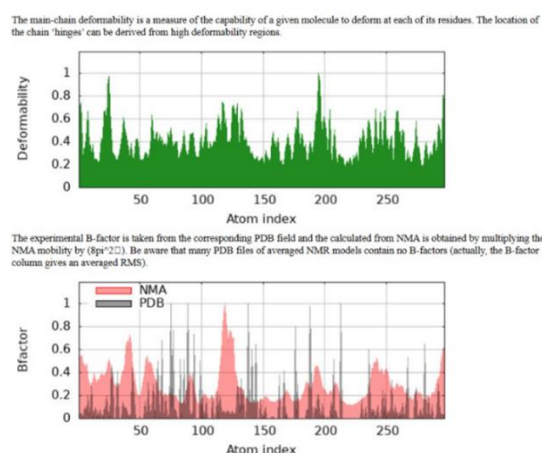


Figure 3.7: Structural Flexibility and Mobility Comparison via Deformability and B-factor Analysis

The two plots presented in the image are about the dynamical properties of the protein with respect to main-chain deformability and B-factor analysis, which confer the protein structure with flexibility and movement. The top graph, 'Deformability,' integrates the y-axis as deformability for individual residues and the x-axis with the atom index along the protein chain. This study is an important tool designed to view flexible regions popularly called hinges that are required for dynamic behavior and function of proteins. The peaks observed on the graph are the residues that seem to have higher deformability and suggest regions able to undertake very significant structural changes. These flexible sites tend to be associated with conformational transitions

fundamental for ligand binding, enzymatic activity, protein-protein contacts.

The lower graph shows the comparison between the experimental B-factors (from the Protein Data Bank) and derived NMA-based B-factors. The B-factor, also known as the Debye-Waller factor or temperature factor, is the measure of atomic displacement or flexibility of specific residues within the structure. Here, the red shaded area corresponds to NMA-derived B-factors, while the thin black line corresponds with experimental PDB B-factors. Patterns between both will suggest a good agreement between computational prediction and experiment. Regions where both peak and broaden will indicate those with higher mobility, which may also co-occur with the locations identified as deformable in the first graph.

The above analysis demonstrates that certain more flexible and dynamic regions are present in proteins according to deformability and B-factor analysis. Most importantly, this information provides a

certain perspective to functional sites in proteins and some guidance regarding making mutations, potential drug designs, and molecular docking studies.

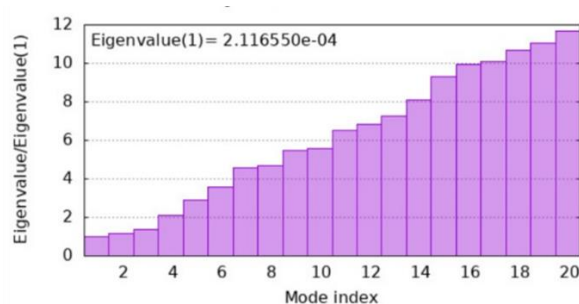


Figure:3.8: Normalized Eigenvalue Spectrum of Top 20 Modes

The picture is a bar graph that shows the distribution of eigenvalues for different mode indices, typical for analyses like NMA or PCA of molecular dynamics or structural bioinformatics. The x-axis refers to the mode index subdivided from 1 to 20, while the y-axis is the normalized eigenvalues, where one eigenvalue is divided by the first eigenvalue. The first eigenvalue is given as 2.11655×10^{-4} . The graph shows that eigenvalues tend to increase steadily, meaning that higher mode indices represent higher vibrational frequencies or stiffer modes, meaning that lower modes (associated with smaller

eigenvalues) correspond to larger, collective, and more flexible motions of the molecular structure relevant for biological functions, such as domain movements. Higher modes are more localized and rigid motions. These eigenvalue spectra aid greatly in identifying the dominant modes of motion useful to guide coarse-grained simulations and understand the biomolecules' structural flexibility and dynamics, which are extremely important for applications in drug design, protein engineering, and conformational analysis.

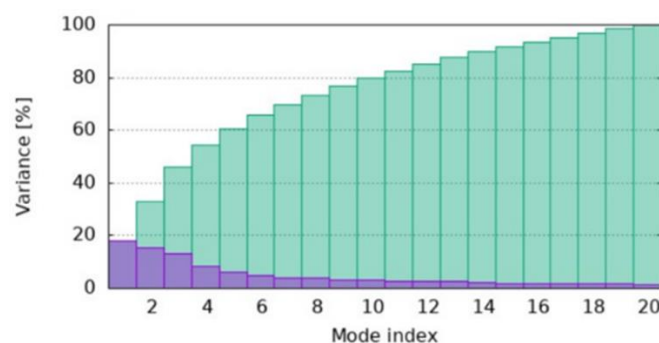


Figure 3.9: Variance Explained by Principal Components

The illustrated image is a bar graph, which represents the distribution of variance explained by different mode indices, typically generated as part of a principal component analysis (PCA) or normal mode analysis (NMA) of dimension-producing datasets in molecular dynamics simulations or between structure data. On the x-axis are modes numbered from one to twenty while on the y-axis is the variance explained in percentage given on the scale. Each bar has two sections: the purple part shows the distance which each mode individually contributes to variance, and the summed up green part indicates the total variance captured up to that mode. When looking at both modes together, it indicates how much variance is explained by each mode on its own and, as more of them is added, the total variance being captured. It is clear now that most of the total variance

can be attributed to the initial few modes, particularly the first five modes, which would imply that the said modes capture the most dominant motions or conformational changes in the system. Although further in the mode index, per mode contribution decreases, cumulative variance draws steadily closer to 100%, meaning that higher order modes contribute less per mode but fully describe the variability of system.

Such types of plot diagram are essentially used to find the top modes that account for most of the canonical behavior of the system; this is particularly important for the purpose of dimensionality reduction in simulating biologically relevant motions or for targeted drug design, wherein one concentrates only on the most significant dynamic features of a protein or molecular complex.

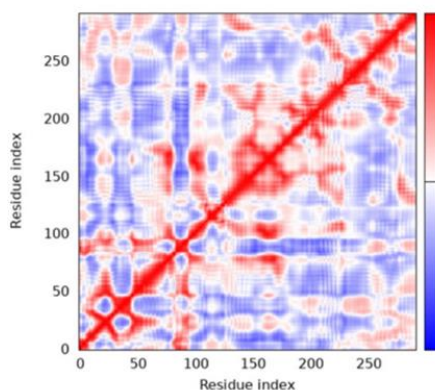


Figure 3.10: Cross Residue -Correlation Map from Molecular Dynamics Simulation

It is a picture of the cross-correlation matrix plot functionally representing residue dynamic behaviour in proteins subjected to molecular dynamics simulation or Normal Mode Analysis (NMA). Both x- and y-axis show residue indices from all protein amino acids. Every point in this matrix denotes the correlated movement degree between a pair of residues, while the colour scale denotes how strong and in what direction its correlation is: red means there is positive correlation in movements (the two residues are moving in the same direction), blue means negative correlation in movements (the two residues are moving opposite directions), and white signifies little to no correlation. The very strong amid deep red

diagonal lines reflect a perfect self-correlation of each residue with itself. Red is off-diagonal and blue pins down the dynamic coupling of residues moving together or in opposition. This could probably denote the flexible domains, hinge regions, or possibly long-range allosteric interactions within the protein. This analysis is paramount in giving insights regarding the protein dynamics, offers signs about its functional domains, and even paths through which the different regions of the molecule communicate. All the said attributes hold potentials that are far-reaching in protein engineering and allosteric drug design.

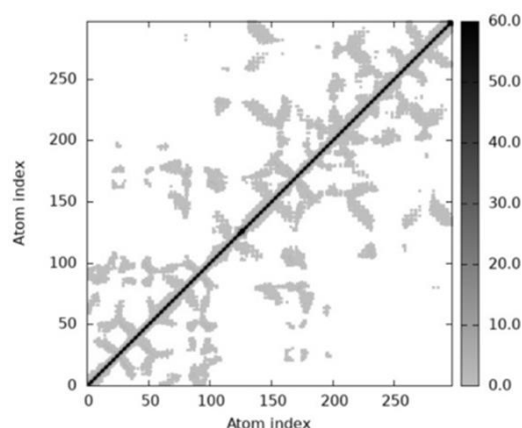


Figure 3.11: Atomic Distance Matrix of Molecular Structure

Basically that image illustrates a distance matrix plot of the type usually employed in structural biology and molecular dynamics in which pairwise distances between atoms are drawn up, being, for instance, those in a molecule like a protein. In this way, the atom indices are present both in the x and in the y axis, and each cell in the matrix indicates the distance between a pair of atoms. The grayscale colour bar depends on the right-hand extremes from 0 up to 60 Å, where light shades indicate short distances while dark shades indicate a long distance. The dark solid line across the diagonal represents the expected zero distance from atom to itself along the main diagonal. Off-diagonal spots illustrate different spatial relationships for different atoms, with patches of lighter gray indicating regions of close packing for the molecule, which may correspond to folded structural elements, such as alpha-helices or beta-sheets. The overall spread of gray intensities can also aid in understanding the three-dimensional

arrangement of the molecule within compact or extended regions. The analysis presented here aids in insights regarding the structural arrangement of a biomolecule, recognition of folding patterns, and aiding comparison between different conformational states; hence, it is a mainstay in the study of protein modeling and dynamics.

4. Conclusion

This research established an effective multi-layered computational framework to pinpoint potential drugs among those already approved by the FDA for use as inhibitors of the FGFR3 K650E mutant, a concerning well-known oncogenic variant implicated in aggressive bladder cancer. Structural refinement, virtual screening, and molecular dynamics simulations were performed to provide insight into binding affinity and structural dynamics.

The 3D crystal structure of the FGFR3 K650E mutant (PDB ID: 4K33) was improved by PDB-REDO and validated by several other means that testify for the better-structured quality. Prominent refinement parameters, such as R-free, were decreased from 0.2351 to 0.2214, bond length RMS Z-score changed from 0.853 to 0.368, and normality in the Ramachandran plot was increased from 52% to 60%, all indicating a reliable model amenable for downstream computational studies.

Virtual screening was performed on the DrugRep platform against a library of FDA-approved compounds where several candidates emerged as promising hits with considerable affinities. Irinotecan (DrugBank ID: DB00762) was ranked on top with a docking score of -10.8 kcal/mol, followed by Eltrombopag (-10.4), Lurasidone (-10.2), and Saquinavir (-10.1). The top 100 compounds had docking scores from -10.8 to -7.0, more than 30 compounds score above -9.5, and indicating high feasibility for repurposing.

The top hits were also analyzed for Normal Mode Analysis (NMA) using the iMODS server. This has led to the acquisition of important information about the protein-ligand interface, conformational flexibility, and movement profiles of the protein-ligand complexes. The deformability plot and B-factor analysis exhibited correlations in flexibility profile, pin-pointing hinge

regions of possible functional motion. The normalized eigenvalue spectrum also pointed out that the first few modes, especially the first mode with eigenvalue 2.11655×10^{-4} , dominated the motion which represented large-scale biologically pertinent dynamics.

The principal component analysis (PCA) confirmed that the first five modes explain a large amount of conformational variance, while dynamic cross-correlation maps highlighted inter-residue communication patterns and flexible domains, with validation of the distance matrix underlining structural compactness and spatial relationships among the protein.

The top reasonable binding pocket, C1, showed a volume of 1799 Å³, thus furnishing larger or complex ligands with accommodation in binding. Overlays of the protein-ligand interactions also revealed strong electrostatic and hydrogen-bonding interactions with residues critical to the binding, such as R655, E575, and K508, thus adding further credibility to the biological plausibility of these binding modes.

In conclusion, this study presents strong computational evidence predicting that several FDA-approved drugs, especially Irinotecan, Eltrombopag, and Lurasidone, could warrant repurposing as inhibitors of the FGFR3 K650E mutant. These findings

create a robust framework for future in vitro validation.

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